AI-Based Land Use Classification of the Delhi-NCR Airshed Using Satellite Imagery

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1 Introduction

Air pollution in the Delhi-NCR region is influenced by surrounding land-use patterns such as urban development, vegetation, and agricultural fields. Understanding land-use distribution through satellite imagery can support environmental monitoring and policy planning.

This project develops a complete geospatial AI pipeline to classify land-use categories across the Delhi-NCR airshed using Sentinel-2 satellite image patches and ESA WorldCover 2021 land-cover data. The pipeline integrates spatial filtering, raster-based label construction, and deep learning models for land-use classification.

The main objectives of this project are:

- Perform spatial reasoning and filtering of satellite imagery.
- Construct land-use labels using ESA WorldCover raster data.
- Train deep learning models for land-use classification.
- Evaluate model performance using accuracy, F1-score, and confusion matrices.

2 Dataset Description

The following datasets were used:

- Sentinel-2 RGB image patches (128×128 pixels, 10m resolution)
- ESA WorldCover 2021 raster dataset (`land_cover.tif`)
- Delhi-NCR boundary shapefile (EPSG:4326)
- Delhi Airshed boundary shapefile

Each satellite tile represents an approximate area of:

$$128 \times 10m = 1.28km \times 1.28km$$

The ESA WorldCover raster dataset provides land-cover classification codes that were used to generate labels for each satellite image.

3 Q1: Spatial Reasoning and Data Filtering

3.1 Plotting Delhi-NCR with Spatial Grid

The Delhi-NCR shapefile was visualized using GeoPandas and Matplotlib. A uniform grid of approximately 60×60 km was overlaid to analyze spatial coverage.

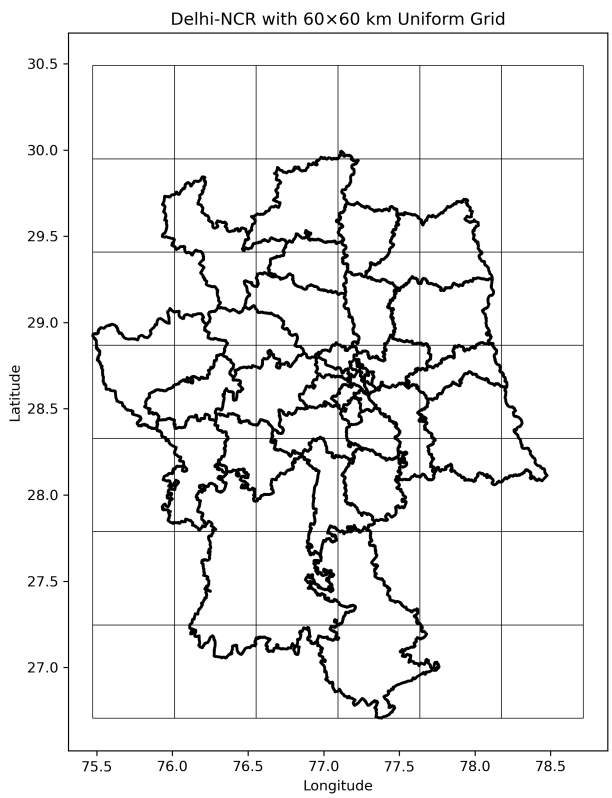

Figure 1: Delhi-NCR boundary with 60×60 km uniform grid

3.2 Filtering Satellite Images

Satellite images are named using their center coordinates:

`latitude_longitude.png`

These coordinates were extracted and converted into geographic points. A point-in-polygon filtering operation was applied to retain only images whose centers lie inside the Delhi-NCR boundary.

3.3 Image Count

- Total satellite images before filtering: 9216
- Images within Delhi-NCR: 8015

4 Q2: Label Construction and Dataset Preparation

4.1 Extracting Land-Cover Patches

For each satellite image:

1. The center coordinate was extracted from the filename.
2. The corresponding pixel location was obtained in the `land_cover.tif` raster.
3. A 128×128 raster patch centered at the coordinate was extracted.

4.2 Assigning Dominant Land-Cover Label

The dominant land-cover class was determined by computing the mode of pixel values within the extracted raster patch.

4.3 Mapping ESA Codes to Simplified Categories

ESA WorldCover class codes were mapped to simplified land-use categories:

ESA Code	Land Use Category
50	Built-up
40	Cropland
10,20,30	Vegetation
80	Water
Others	Others

4.4 Train-Test Split and Class Distribution

The dataset was divided using a 60/40 train-test split.

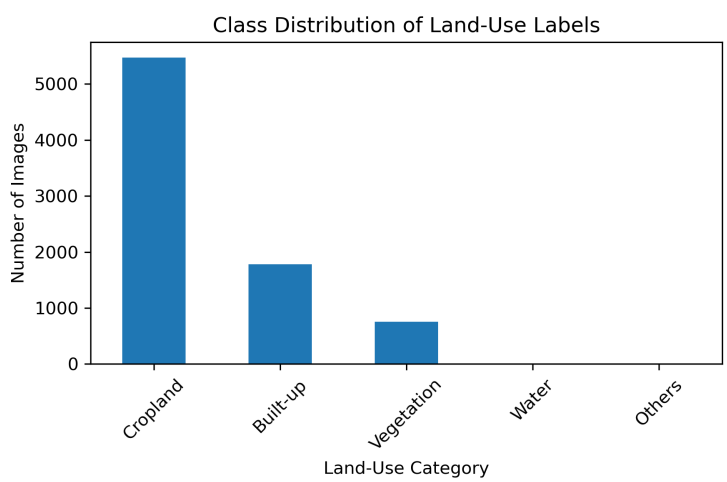

Figure 2: Distribution of land-use classes

The dataset shows a strong imbalance where Cropland is the dominant class.

4.5 Spatial Distribution of Labels

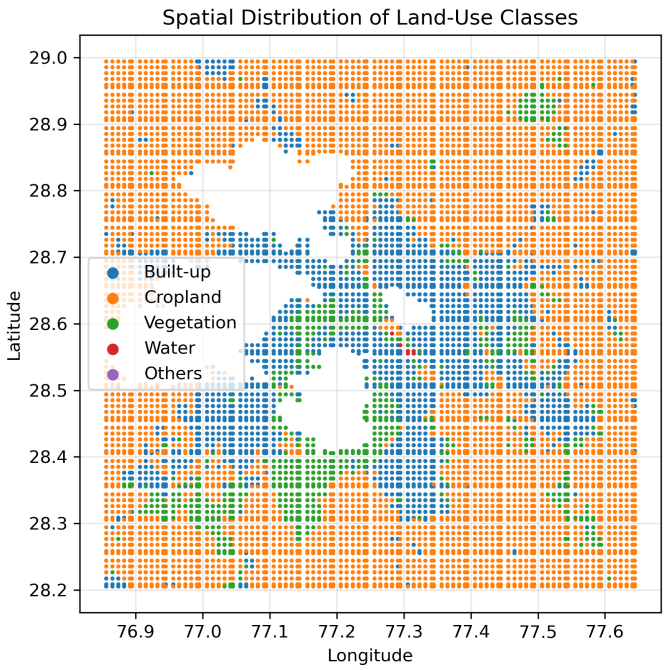

Figure 3: Spatial distribution of land-use classes across Delhi-NCR

5 Q3: Model Training and Supervised Evaluation

5.1 Baseline CNN Model

A custom Convolutional Neural Network (CNN) was implemented consisting of:

- Two convolution layers
- ReLU activation
- MaxPooling layers
- Fully connected classification layers

The CNN was trained using cross-entropy loss.

5.2 CNN Model Evaluation

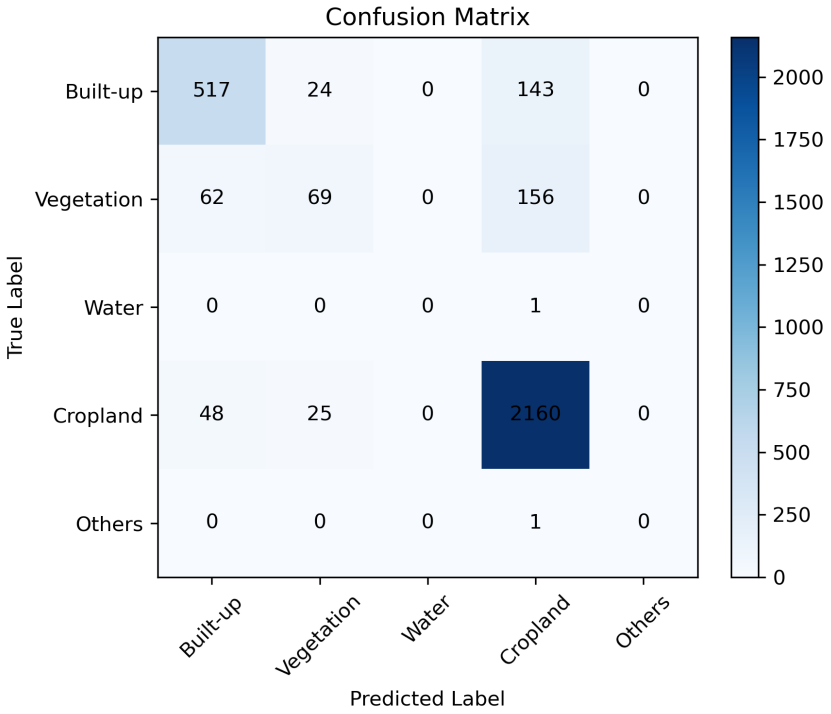

Figure 4: Confusion matrix for baseline CNN model

The CNN model correctly classifies many Cropland and Built-up areas but shows confusion between Vegetation and Cropland due to similar visual textures.

5.3 Improved Model: ResNet18 with Class Balancing

To improve performance, a pretrained ResNet18 architecture was used. Transfer learning enables the model to leverage features learned from large-scale image datasets.

To address dataset imbalance, class-weighted cross-entropy loss was used so that minority classes receive higher importance during training.

5.4 ResNet18 Model Evaluation

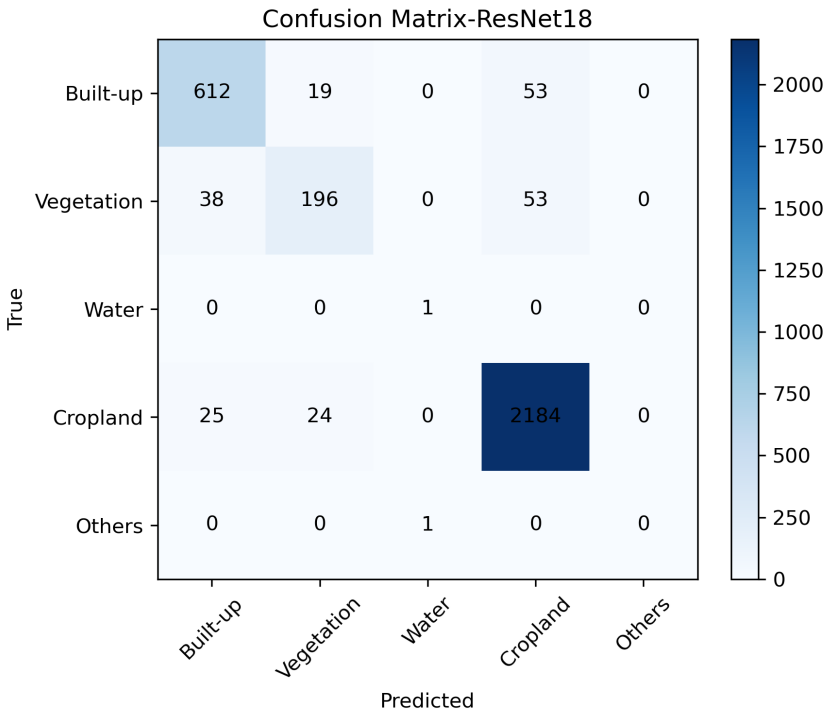

Figure 5: Confusion matrix for ResNet18 model

The ResNet18 model significantly reduces misclassification errors and improves classification performance.

5.5 Model Performance Comparison

Model	Accuracy	F1-score
Baseline CNN	85.65%	0.84
ResNet18 + Class Weights	93.36%	0.93

The improved model achieves higher accuracy and better class balance.

5.6 Prediction Visualization

To better understand model behavior, examples of correct and incorrect predictions were visualized.

Correct predictions typically correspond to areas with clear land-use characteristics such as dense urban regions or large agricultural fields. Incorrect predictions mainly occur between Vegetation and Cropland classes due to similar visual patterns in satellite imagery.

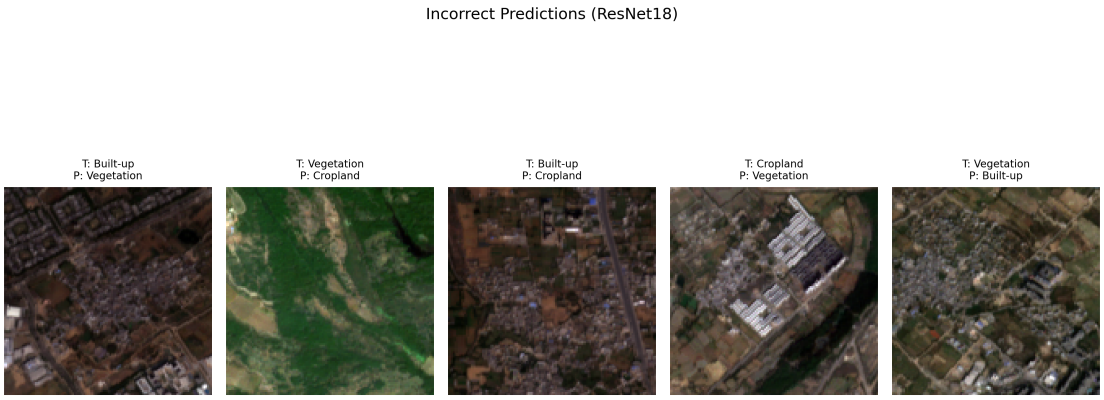

Figure 6: Examples of correctly classified satellite images

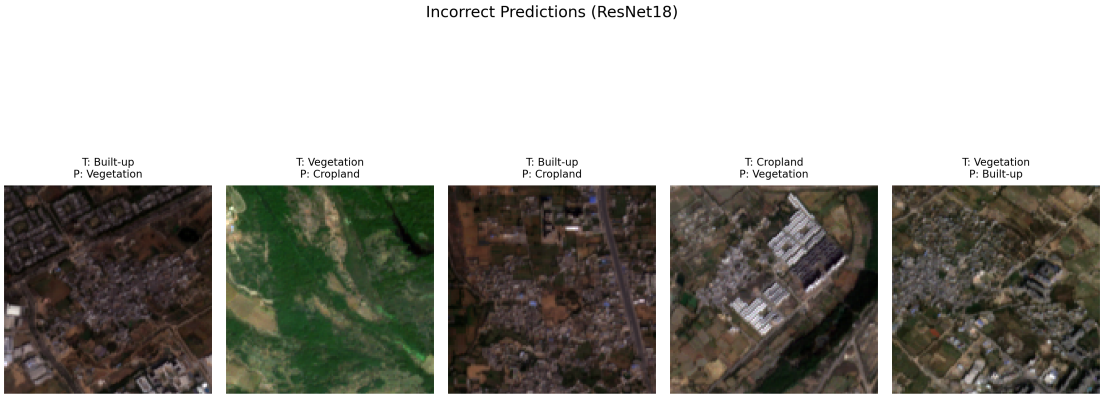

Figure 7: Examples of incorrectly classified satellite images

6 Discussion

The results demonstrate that transfer learning significantly improves remote sensing classification performance. The pretrained ResNet18 model captures deeper spatial features compared to the shallow CNN.

Applying class-weighted loss also helps mitigate dataset imbalance and improves prediction of underrepresented classes.

Remaining classification errors are primarily caused by spectral similarity between vegetation and cropland regions.

7 Conclusion

This project demonstrates a complete geospatial AI workflow integrating spatial filtering, automated label generation from raster data, and deep learning classification.

The ResNet18 model achieved an accuracy of 93.36% and a weighted F1-score of 0.93, demonstrating strong performance in land-use classification for the Delhi-NCR region.

8 Use of AI Tools and External Resources

During the development of this project, AI-assisted tools such as ChatGPT were used for conceptual clarification, debugging assistance, improving code structure, and refining written explanations. These tools were used as learning support to better understand geospatial data processing, PyTorch-based deep learning workflows, and satellite image analysis.

Online documentation and open-source resources were also referenced for understanding libraries such as GeoPandas, Rasterio, Matplotlib, and PyTorch.

All code implementations, experiments, and analysis were executed manually after fully understanding the concepts involved. AI tools were used only as learning assistance, and the final implementation reflects a clear understanding of the techniques applied in this project.